DESIGN THINKING

& FABRICATION LABORATORY

About this Course

Design Thinking is a human-centred, iterative approach to innovation that puts *people first* and turns real problems into workable solutions.

- Blends **empathy**, **creativity** and **rationality**.
- Moves from *understanding people* to *building and testing* tangible solutions.
- The **Fabrication Laboratory** turns ideas into physical prototypes using digital fabrication tools.

Course structure: five units of 15 hours each, combining theory with hands-on making.

Learning outcomes

By the end of the course you will be able to:

- Understand how people learn and empathise.
- Apply the 5-stage design-thinking process.
- Design, prototype and test products.
- Map customer journeys and service blueprints.
- Build solutions in a fabrication lab.

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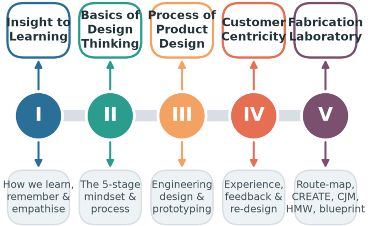
Course Roadmap

Course Roadmap — Five Units, One Journey



15 hours per unit · 75 hours total · theory + hands-on fabrication

Five units, one continuous journey from insight to fabrication.

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Outline

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I

An Insight to Learning

Learning, memory, emotion & empathy

Understanding the Learning Process

Learning is the process of acquiring **knowledge**, **skills** and **attitudes** through experience, study or teaching.

- **Cognitive** — thinking, understanding, problem solving.
- **Affective** — emotions, values, motivation.
- **Psychomotor** — physical / hands-on skills.

Effective designers are *lifelong learners*: they observe, reflect, conceptualise and experiment — exactly the loop Kolb described.

Why it matters for design

Understanding *how humans learn* helps us design products, instructions and experiences that people can pick up quickly and use intuitively.

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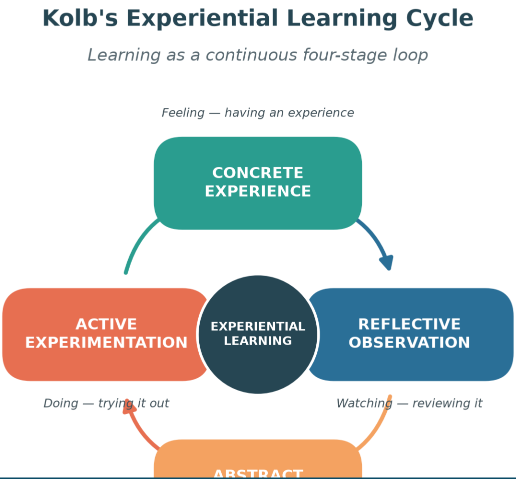
Kolb's Experiential Learning Cycle



Learning is a **continuous cycle** of four stages:

1. **Concrete Experience** — do / feel.
2. **Reflective Observation** — review.
3. **Abstract Conceptualisation** — conclude.
4. **Active Experimentation** — plan & try.

Effective learning happens when we travel *all the way round* the loop — repeatedly.

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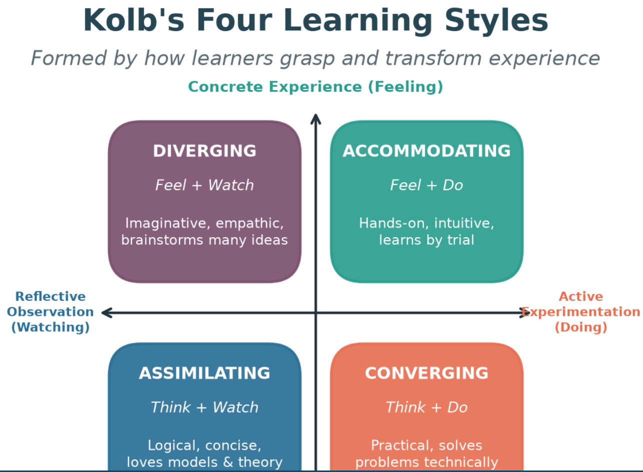
Kolb’s Four Learning Styles



Combining *how we grasp* and *how we transform* experience gives four styles:

- **Diverging** — imaginative, many ideas.
- **Accommodating** — hands-on, intuitive.
- **Assimilating** — logical, loves theory.
- **Converging** — practical problem-solver.

Teams work best when *all four styles* are present.

Assessing & Interpreting Learning Styles

How we assess

- **Learning Style Inventory (LSI)** questionnaires.
- Self-reflection journals.
- Observation of behaviour in tasks.

How we interpret

- No style is *better* — each has strengths.
- Match **teaching methods** to styles.
- Encourage learners to stretch into weaker modes.

Application with peers

In group projects, deliberately mix styles so ideation, analysis, building and testing are all covered.

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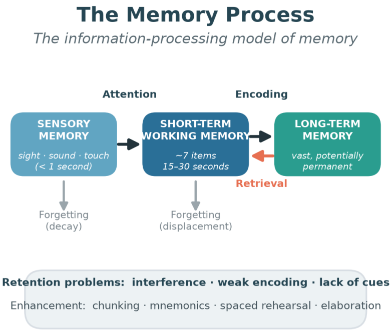
The Memory Process

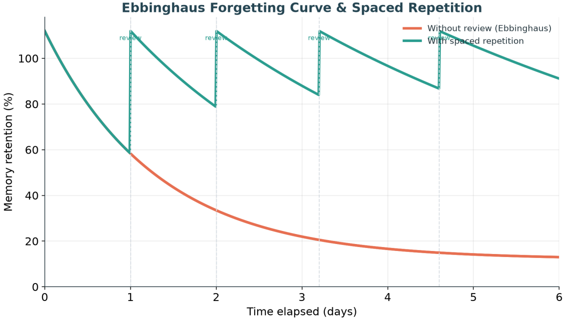


The information-processing model: sensory → short-term → long-term memory.

Problems in Retention

Why do we forget?

- **Decay** — traces fade without use.
- **Interference** — old & new memories clash.
- **Retrieval failure** — no cue to recall.
- **Weak encoding** — shallow processing.



The **Ebbinghaus curve** shows rapid forgetting — but each review flattens the curve.

Memory Enhancement Techniques

- **Chunking** — group items into meaningful units.
- **Mnemonics** — acronyms, rhymes, imagery.
- **Spaced repetition** — review at intervals.
- **Elaboration** — connect to prior knowledge.
- **Dual coding** — words + visuals together.
- **Active recall** — test yourself, don't re-read.
- **Method of loci** — memory "palace".
- **Sleep & rest** — consolidate memories.

Design tip

Good product design *reduces the memory load* on the user — clear labels, consistent layouts and helpful cues.

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Emotions, Experience & Expression

Emotions shape how people **experience** and **remember** products.

- **Emotion** — an internal feeling (joy, frustration, trust).
- **Experience** — the felt journey of using something.
- **Expression** — how emotion is shown & communicated.

Peak emotions (delight or annoyance) leave the strongest memories — the *peak-end rule*.

Emotional design (Norman)

- **Visceral** — first look & feel.
- **Behavioural** — pleasure of use.
- **Reflective** — meaning & self-image.

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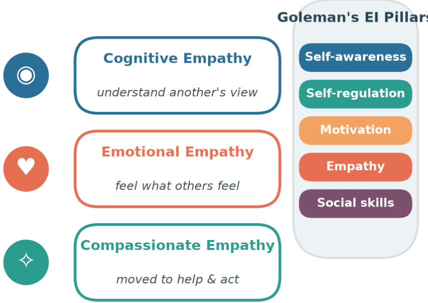
Empathy & Emotional Intelligence

Emotions, Empathy & Emotional Intelligence

Understanding emotions is the gateway to human-centred design



Assessing empathy with peers → empathy maps, shadowing, and active-listening interviews

Empathy — the ability to understand and share another's feelings — is the *heart of design thinking*. **Assessing empathy with peers**

- Empathy maps (says / thinks / does / feels).
- Shadowing & observation.
- Active-listening interviews.

II

Basics of Design Thinking

The mindset, the process, the tools

What is Design Thinking?

Definition

A **human-centred**, iterative problem-solving approach that seeks to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test.

Need for design thinking

- Problems are **complex & ill-defined**.
- Users' real needs are often *hidden*.
- Reduces the risk of building the wrong thing.

Objectives

- Keep the **user at the centre**.
- Encourage **creativity & collaboration**.
- **Fail early**, learn cheaply.
- Turn insight into tangible value.

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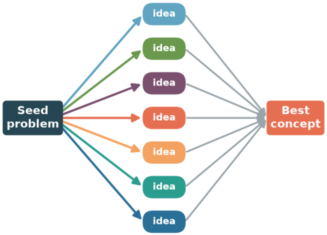
Concepts & Brainstorming

Brainstorming — Divergent then Convergent Thinking

Rules: quantity over quality · build on others · one conversation · stay visual



DIVERGE — generate many ideas, defer judgement, go wild

CONVERGE — group, evaluate & select

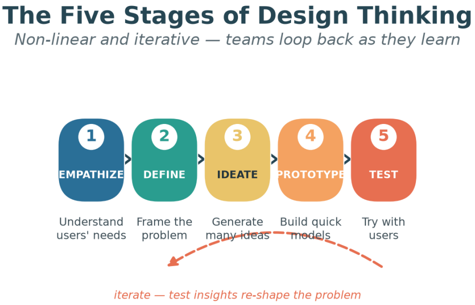
Creative work alternates two modes:

- **Diverge** — generate many ideas, defer judgement.
- **Converge** — group, evaluate & select.

Brainstorming rules

- Go for quantity.
- Build on others' ideas.
- One conversation at a time.
- Stay visual; welcome wild ideas.

The Five Stages of Design Thinking



Empathize → Define → Ideate → Prototype → Test — iterative, not linear.

Stages in Detail (1–3)

[1. Empathize] Observe and engage with users; conduct interviews and immerse yourself in their world. *Example:* shadowing commuters to learn how they use a ticket machine.

[2. Define] Synthesise observations into a clear, user-centred **problem statement** (a Point-of-View). *Example:* "Busy commuters need a faster way to buy tickets because queues cause stress."

[3. Ideate] Generate a wide range of ideas through brainstorming, sketching and mind-mapping — *quantity over quality* first.

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Stages in Detail (4–5)

[4. Prototype] Build quick, inexpensive, scaled-down versions to explore ideas. *Example:* a paper mock-up of the ticket-machine screen.

[5. Test] Try prototypes with real users, gather feedback and refine. Testing often *re-defines the problem* — sending you back to earlier stages.

Being ingenious

The stages are a *flexible toolkit*, not a rigid recipe. Teams loop, skip and revisit stages as they learn.

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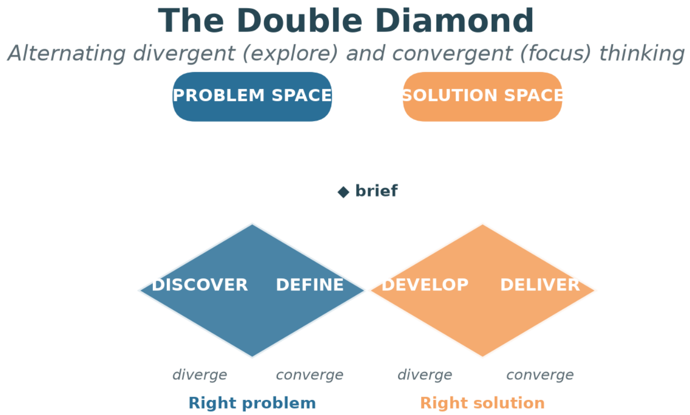
The Double Diamond



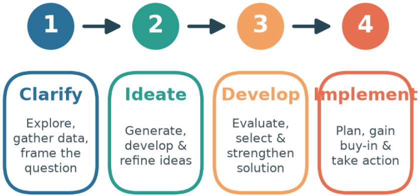
A popular map of the design process:

- **Discover & Define** — find the *right problem*.
- **Develop & Deliver** — build the *right solution*.

Each diamond widens (diverge) then narrows (converge).

Creative Thinking & Problem Solving

Creative Problem-Solving Process



Creative problem solving moves through four stages: **Clarify**, **Ideate**, **Develop** and **Implement**. **Testing creative problem solving** — judge ideas on:

- *Fluency* — number of ideas.
- *Flexibility* — variety of ideas.
- *Originality* — novelty.
- *Elaboration* — detail & refinement.

Testing creative problem solving: judge originality, fluency, flexibility & elaboration of ideas

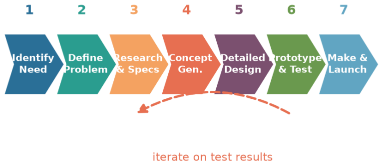
III

Process of Product Design

Engineering design, prototyping & testing

Engineering Product-Design Process

Engineering Product-Design Process



design-thinking approach keeps the user in the loop at every stage

A structured flow from identifying a need to manufacture & launch.

Design-Thinking Approach to Product Design

Traditional engineering

- Requirements defined up front.
- Linear, specification-driven.
- User feedback comes late.

Design-thinking infused

- **User needs** continuously explored.
- **Iterative** — prototype & test early.
- Feedback loops at *every* stage.

Best in class

Great products (e.g. intuitive smartphones, ergonomic tools, well-designed appliances) combine *function*, *form* and *delight* — form follows function *and* feeling.

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What is a Prototype?

A **prototype** is an early, testable model of a product used to *explore ideas* and *learn* before committing to production. **Why prototype?**

- Make ideas **tangible**.
- Test assumptions cheaply.
- Communicate with stakeholders.
- Fail fast, improve faster.

Types of prototype

- *Low-fidelity* — sketches, paper, cardboard.
- *Mid-fidelity* — wireframes, models.
- *High-fidelity* — 3-D printed, functional.

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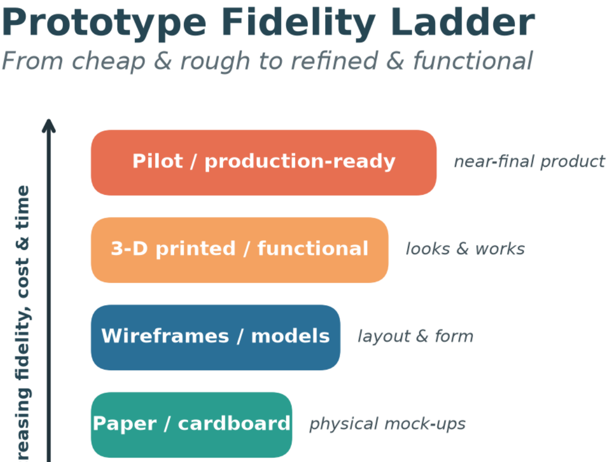
Prototype Fidelity Ladder

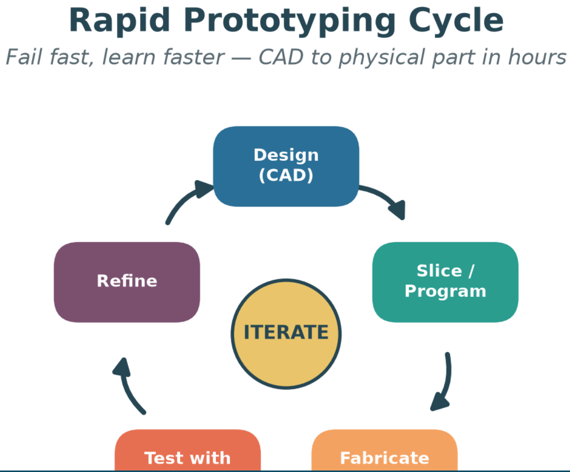


Start low and cheap, climb only as questions demand.

- Low fidelity answers *"is this the right idea?"*
- High fidelity answers *"does it work & feel right?"*

Each rung costs more time and money — so *learn cheaply first*.

Rapid Prototype Development



Rapid prototyping uses digital fabrication (3-D printing, laser cutting, CNC) to go from **CAD** to a physical part in *hours*. **Testing & sample marketing**

- Test the prototype with a small **test group**.
- Gather reactions before scaling up.
- Refine, then repeat the loop.

IV

Design Thinking & Customer Centricity

Experience, feedback & re-creation

Practical Customer Challenges

Real customers face everyday frustrations that design thinking can solve:

- Products that are **hard to assemble** or use.
- Confusing instructions & interfaces.
- Poor **ergonomics** causing discomfort.
- Long waits and clunky service.
- Features nobody asked for.
- A gap between the *promise* and the *experience*.

Design thinking to the rescue

By empathising with customers we uncover these pains and re-design the product experience around them.

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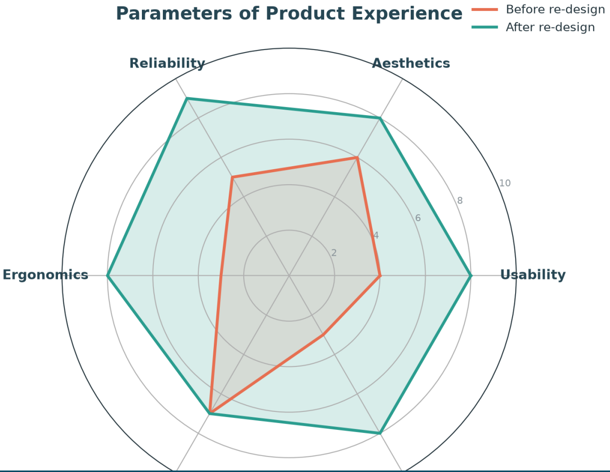
Parameters of Product Experience



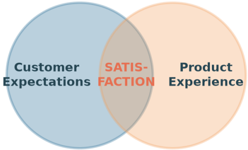
Product experience is multi-dimensional:

- **Usability** & **Reliability**
- **Aesthetics** & **Ergonomics**
- **Affordability** & **Delight**

Thoughtful re-design lifts every axis — turning a merely functional product into one customers *love*.

Aligning Customer Expectations

Aligning Customer Expectations with the Product



Gap too wide → disappointment · aligned → loyalty & advocacy

Satisfaction lives where *expectations* meet *experience*.

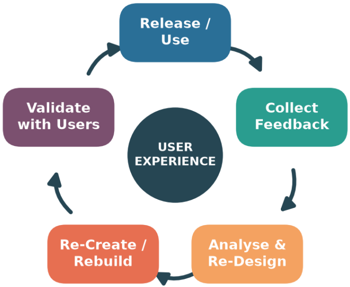
- Gap too wide → disappointment.
- Well aligned → loyalty & advocacy.

Manage expectations honestly *and* raise the experience to close the gap.

Feedback, Re-Design & Re-Create

Feedback → Re-Design → Re-Create Loop

Continuous improvement centred on user experience



Address ergonomic challenges · close the gap between expectation and experience

The **feedback loop** keeps a product improving:

1. Release & use.
2. Collect feedback.
3. Analyse & re-design.
4. Re-create / rebuild.
5. Validate with users.

Focus relentlessly on **user experience** and address ergonomic challenges each cycle.

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User-Focused Design & the Final Product

User-focused design

- Design *with* users, not just for them.
- Fit the product to the human — **ergonomics**.
- Test with real tasks in real contexts.

Toward the final product

- Converge feedback into refinements.
- Validate safety, comfort & usability.
- Ship — then keep listening.

Remember

A "final" product is really the *best current version* — the loop never truly ends.

V

Fabrication Laboratory

The route map: survey to service blueprint

The Fab-Lab Route Map

Fab-Lab Route Map — From Survey to Service Blueprint

A step-by-step recipe for the lab project:

1. Conduct surveys.
2. Identify a problem.
3. Frame a problem statement.
4. Apply the **CREATE** tool.
5. Draw the product / system.
6. Build a Customer Journey Map.
7. Frame 2–3 **HMW** questions.
8. Design a Service Blueprint.

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Surveys, Problems & Problem Statements

Conduct surveys

- Individually or as a group.
- Observe, interview, question.
- Look for **real, felt pains**.

Identify a problem worth solving.

Framing a problem statement

A good statement is **user-centred**, **clear** and **actionable**:

"[User] needs [need] because [insight]."

Avoid baking in a solution — describe the *problem*.

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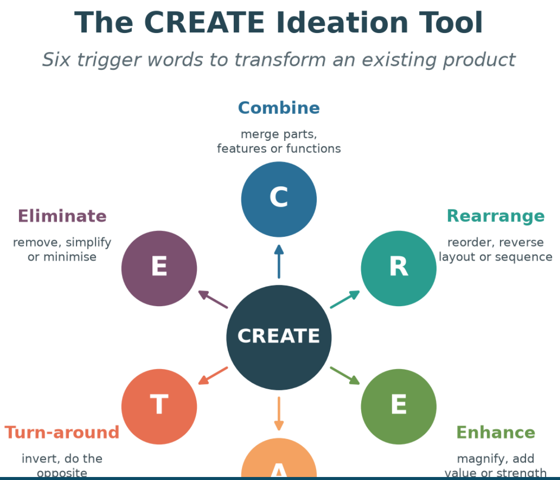
The CREATE Tool



CREATE is a set of *trigger words* to transform an existing product:

- **C**ombine **R**earrange
- **E**nhance **A**dapt
- **T**urn-around **E**liminate

Apply each trigger to your problem, then **draw the product / system** that results.

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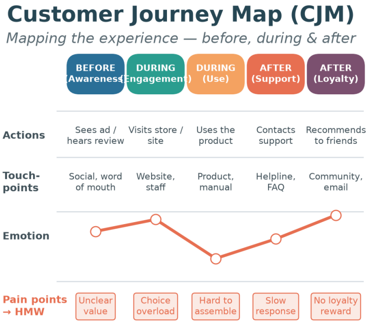
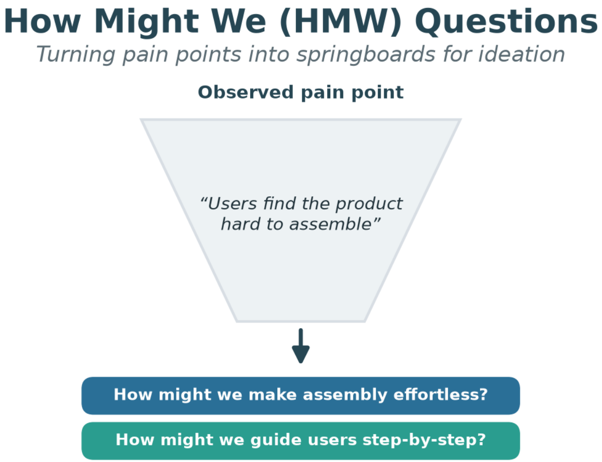
Customer Journey Map (CJM)



Touch-points feed the Service Blueprint · pain points become How-Might-We questions

Map actions, touch-points, emotions & pain points — before, during & after.

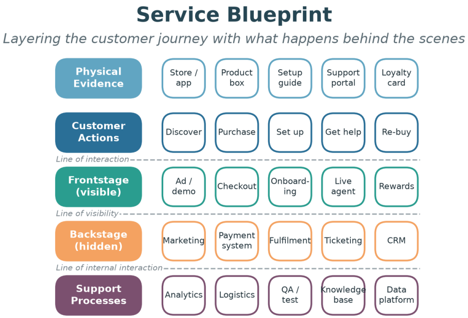
How Might We (HMW) Questions



Turn each **pain point** from the CJM into an optimistic **"How Might We. . . "** question.

- Broad enough to *inspire* ideas.
- Narrow enough to *act* on.
- Frame **2–3** for your mock scenario.

Service Blueprint



Layer the journey with front-stage, back-stage & support processes; identify touch-points.

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Bringing It All Together

- **Unit I** — learn, remember, empathise.
- **Unit II** — the design-thinking process.
- **Unit III** — design & prototype products.
- **Unit IV** — centre on the customer.
- **Unit V** — build & blueprint in the lab.

The golden thread

Every unit returns to one idea: **empathise with people, then iterate** — observe, frame, ideate, prototype, test, and repeat until the experience truly serves the user.

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Design Thinking & Fabrication Laboratory

Empathise → Define → Ideate → Prototype → Test